

History of the Internet & Introduction to Networking





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01

History of the Internet — Overview & Timeline

Internet Evolution Summary

Overview

- Originated from military and academic research in the 1960s
- Started with packet switching and ARPANET
- TCP/IP protocols standardized in 1970s–80s
- NSFNET expanded backbone networks in 1980s–90s
- Commercialization began mid-1990s
- Key innovations: email, DNS, WWW, graphical browsers, broadband, mobile devices
- Internet evolved into an always-on infrastructure supporting communication, commerce and cloud computing



Detailed Timeline

01 Early Innovations (1960s–1980s)

- 1960s: Packet switching concept by Paul Baran, Donald Davies
- 1969: ARPANET connects first 4 nodes
- Early 1970s: Email and early protocols emerge
- 1973–74: TCP designed by Vint Cerf & Bob Kahn
- 1983: ARPANET switches to TCP/IP (“flag day”)
- 1984: DNS introduced for human-readable addresses
- Mid-late 1980s: NSFNET backbone connects universities

02 Web and Commercial Growth (1989–1990s)

- 1989–91: Tim Berners-Lee invents World Wide Web (HTTP, HTML, URLs)
- 1993: Mosaic graphical web browser popularizes multimedia web
- 1990s: Lifting of NSFNET commercial restrictions, rise of ISPs, portals, e-commerce



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03 Modern Internet Developments (1998–2010s)

- 1998: Google founded, search & infrastructure improvements
- Late 1990s–2000s: Broadband replaces dial-up; Web 2.0 emerges
- 2004–2010s: Mobile Internet boom via smartphones, wireless tech (3G/4G)
- 2010s: Cloud computing (AWS, Azure) and IoT expand Internet roles
- IPv6 introduced to expand address space; gradual adoption ongoing



Key People & Organizations

01

Pioneers

- Vint Cerf & Bob Kahn: TCP/IP architects
- Tim Berners-Lee: Web inventor
- Paul Baran, Donald Davies: Packet-switching pioneers

02

Organizations

- DARPA: Funded early development
- NSF: Supported NSFNET academic backbone

Importance of Milestones

Packet switching

efficient, fault-tolerant
communication



TCP/IP

interoperability of
diverse networks



DNS

user-friendly names
essential for Internet growth



World Wide Web + browsers

accessible, multimedia
content



Broadband & mobile

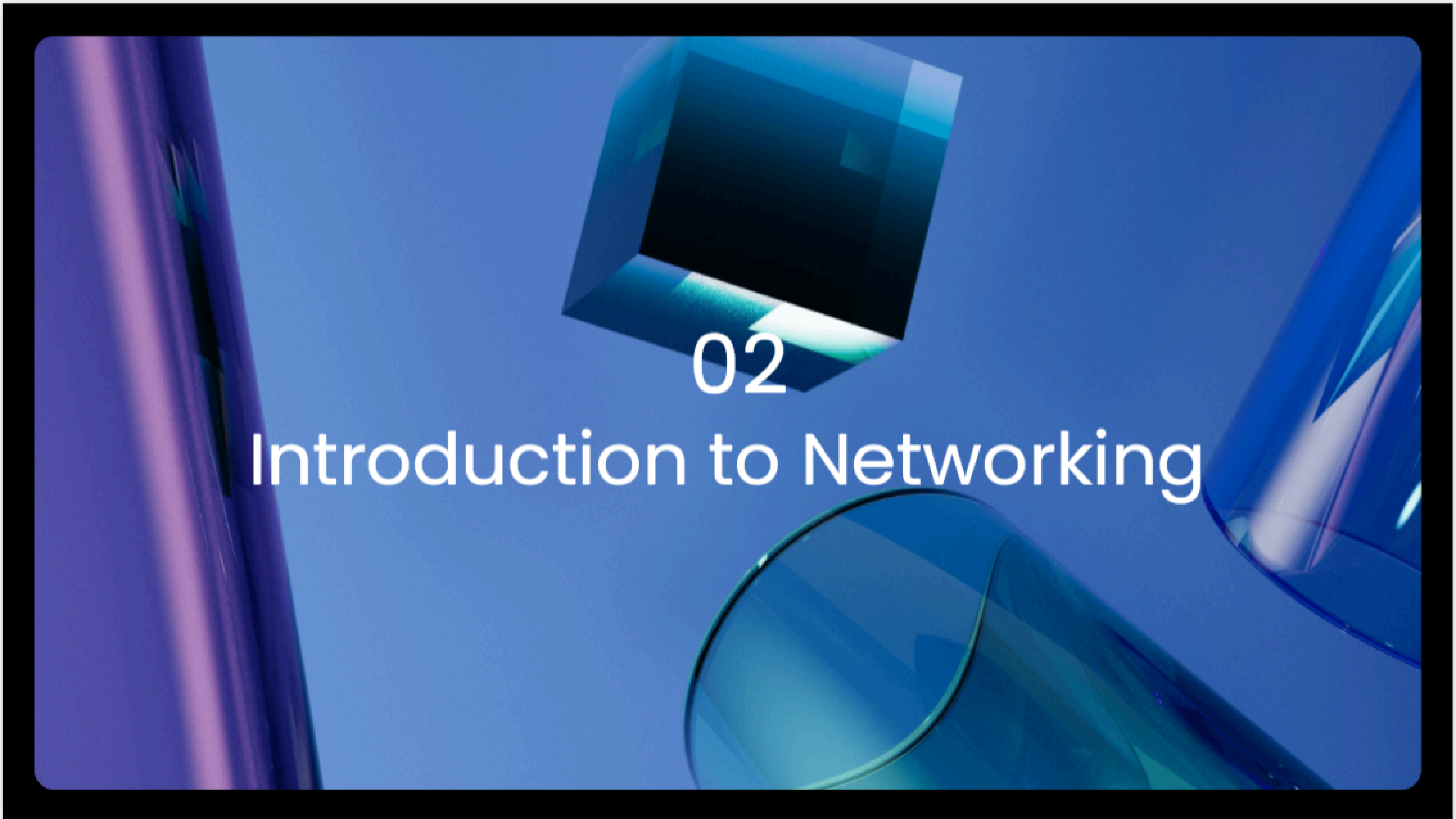
always-on, portable
access



Cloud & CDNs

scalable computing
and content delivery





02

Introduction to Networking

What is a Network?



Definition and Function

- Two or more computers linked to communicate
- Enables resource sharing: data, hardware, software
- Supports global communication and collaboration

Types of Networks

Personal Area Networks (PAN)

Connects nearby personal devices (phones, PDAs) within a few meters
Wired (USB, FireWire) or wireless (Bluetooth, ZigBee)

Local Area Network (LAN)

Small area coverage (homes, offices)
Shares files, printers, applications
Suited for small/medium teams, up to 100 users

Metropolitan Area Network (MAN)

Connects multiple buildings (campuses, companies)
Larger than LAN, smaller than WAN

Storage Area Network (SAN)

Connects servers to storage devices
High-performance data transfer alternative



Types of Networks

Wide Area Network (WAN)

Spans large geographical areas
Example: Internet itself, campus-wide networks

Wireless LAN (WLAN)

Wireless extension of LAN using radio frequency
Provides mobility and connectivity

Other Networks

Intranet: private LAN within organizations
Extranet: secure network connecting internal and external parties



Benefits of Networking

**Information
Sharing**

**Hardware
Sharing**

**Software
Sharing**

**Collaborative
Environment**

**Centralized
Storage**

ADVANTAGES

**Peripheral
Sharing**

Time Saving

Security

**Communi-
cation**

Risks of Network Computing

Threats

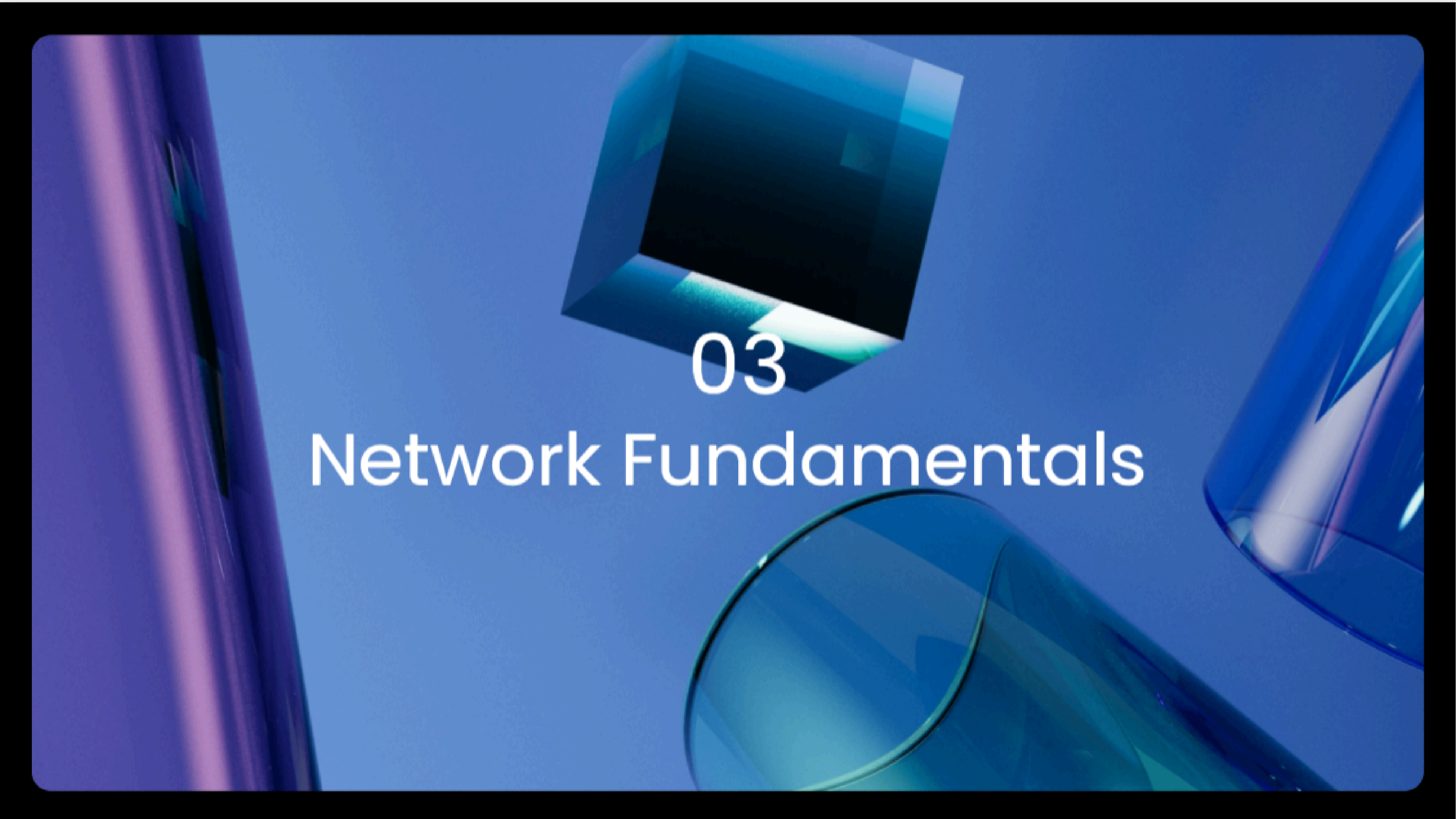


- Equipment malfunctions
- System failures (caused by disasters like floods, storms, or electrical issues)

Security Risk



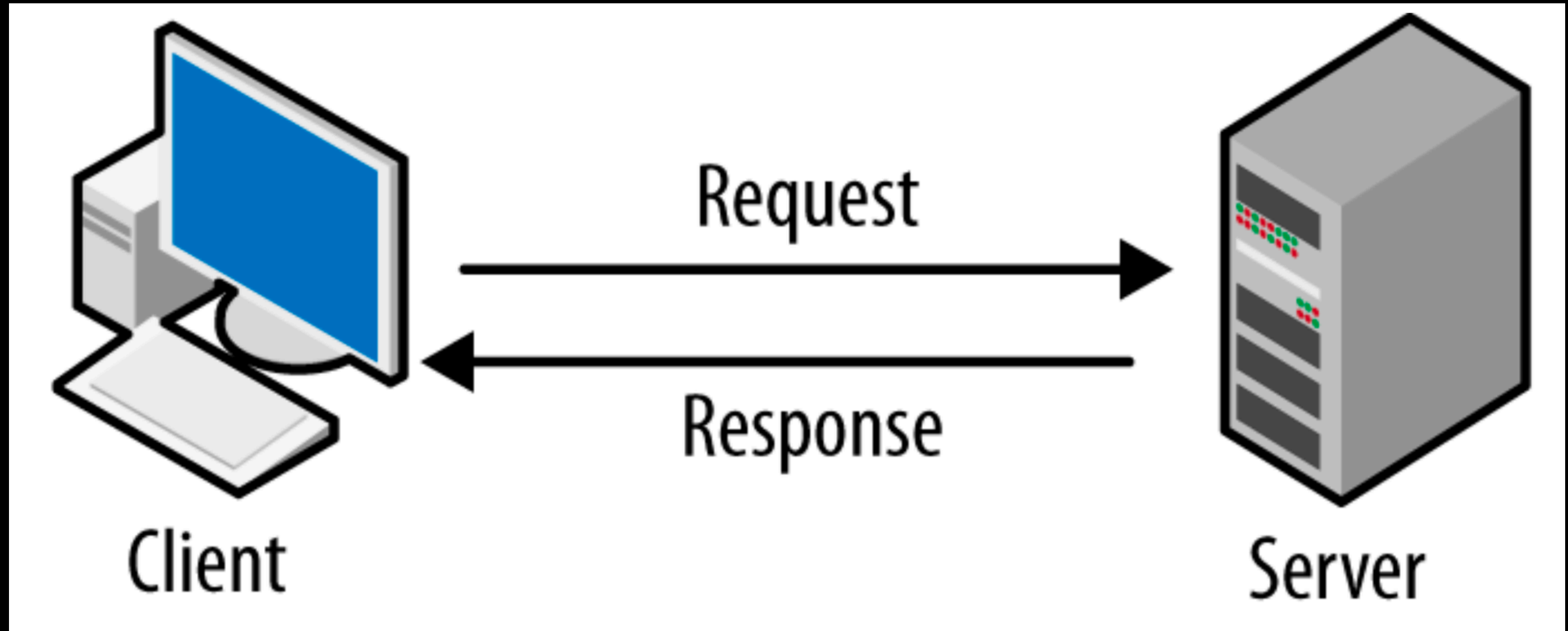
- Hackers
- Virus attacks



03

Network Fundamentals

Network Terminology



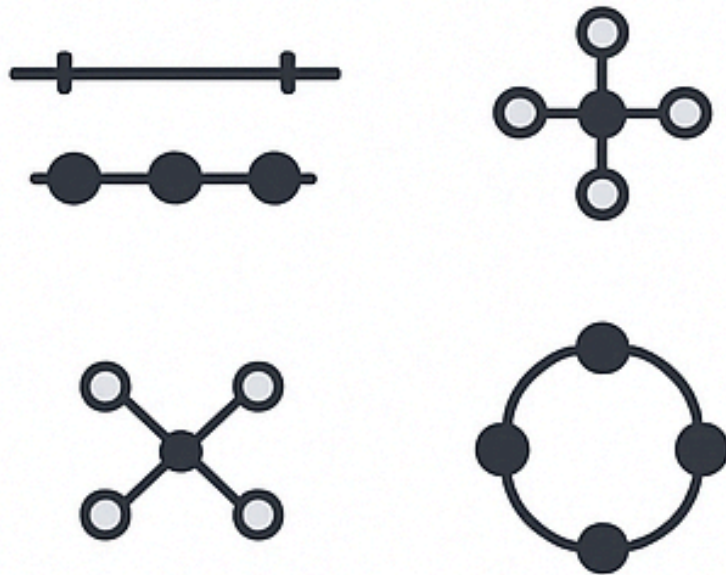
A device or software that requests services from a server

A computer that stores and provides resources, data, and services to the network.

Network Categories

NETWORK TOPOLOGY

The physical or logical arrangement of devices in a network



NETWORK PROTOCOL

Rules and standards for communication between devices

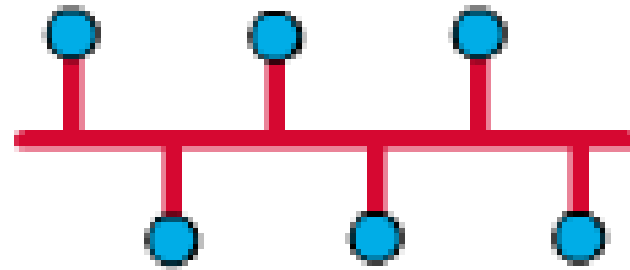
TCP/IP
HTTP FTP
SMTP DNS

NETWORK ARCHITECTURE

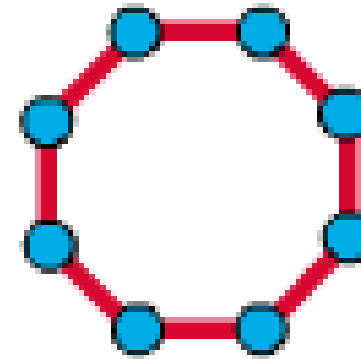
The overall design and structure of a network

- Peer-to-Peer
- Client-Server
- Distributed

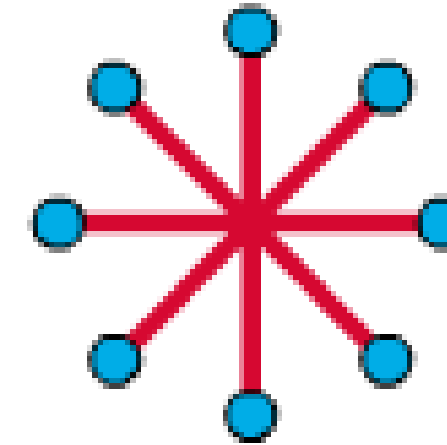
Physical Topology



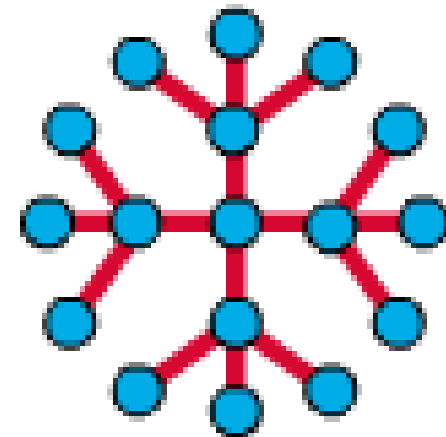
Bus Topology



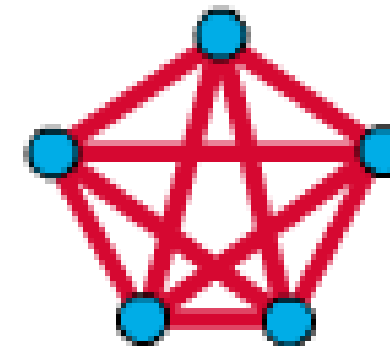
Ring Topology



Star Topology



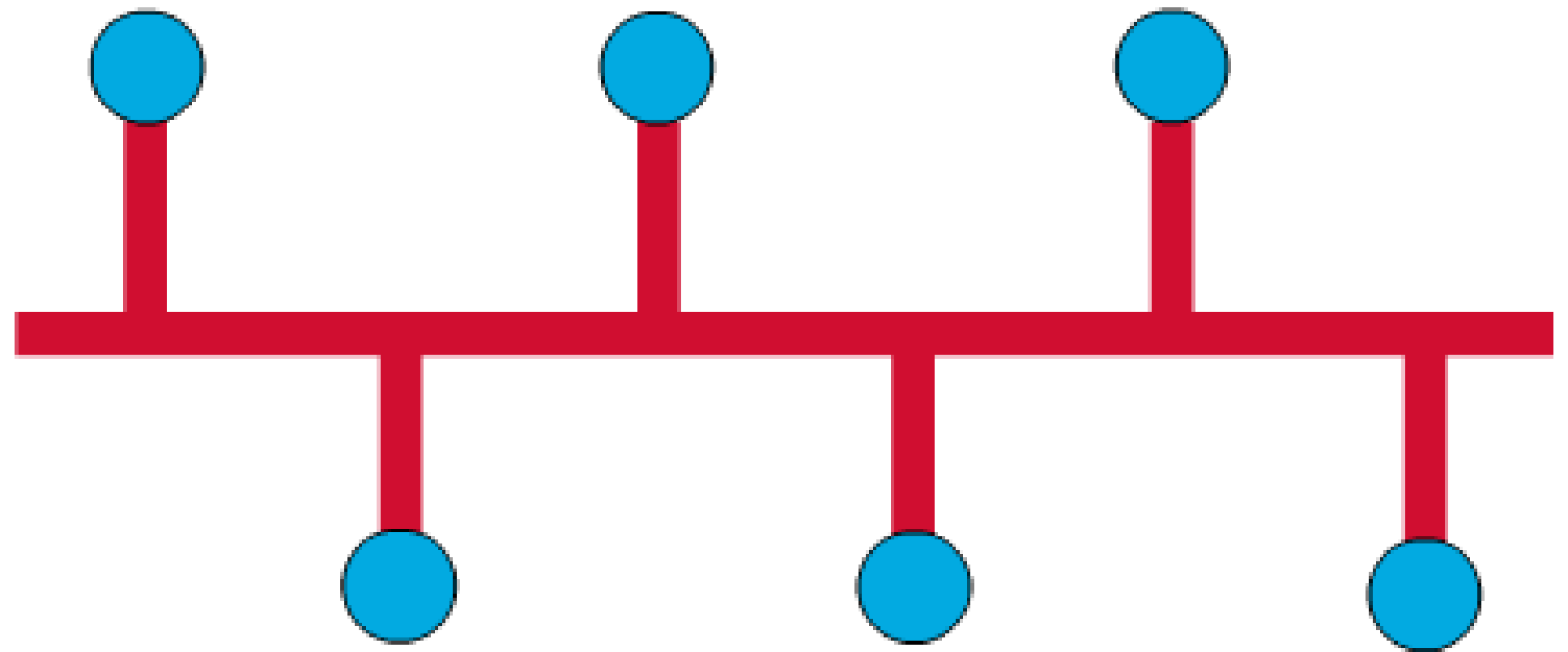
Extended Star
Topology



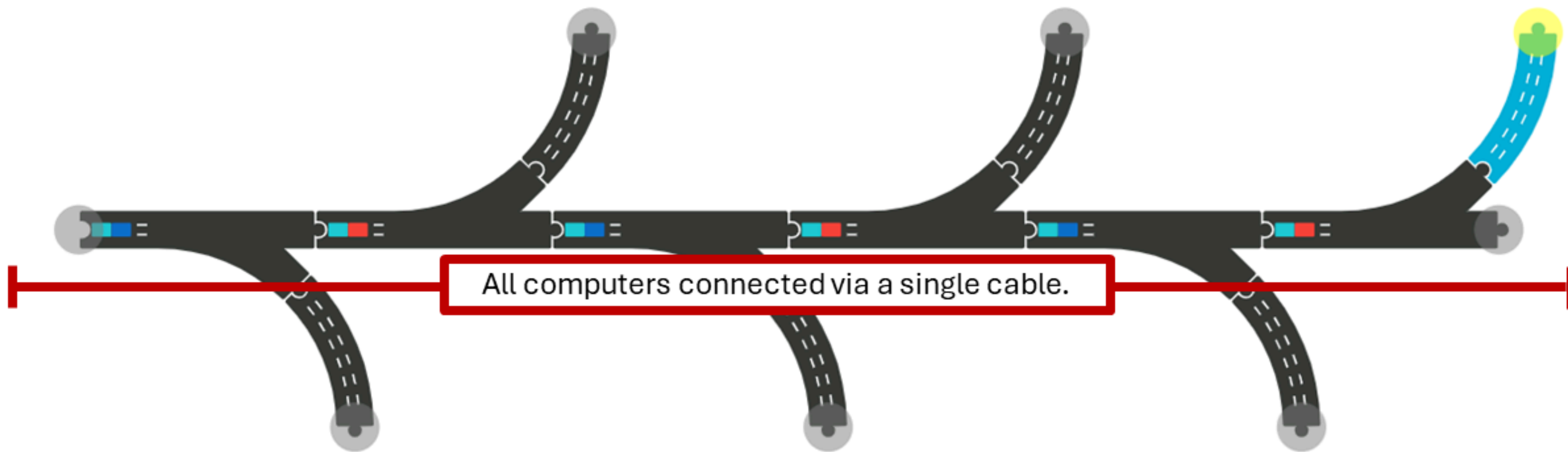
Mesh
Topology

Bus Topology

- Commonly referred to as a linear bus, all the devices on a bus topology are connected by one single cable.



Bus Topology



Advantage of bus topology

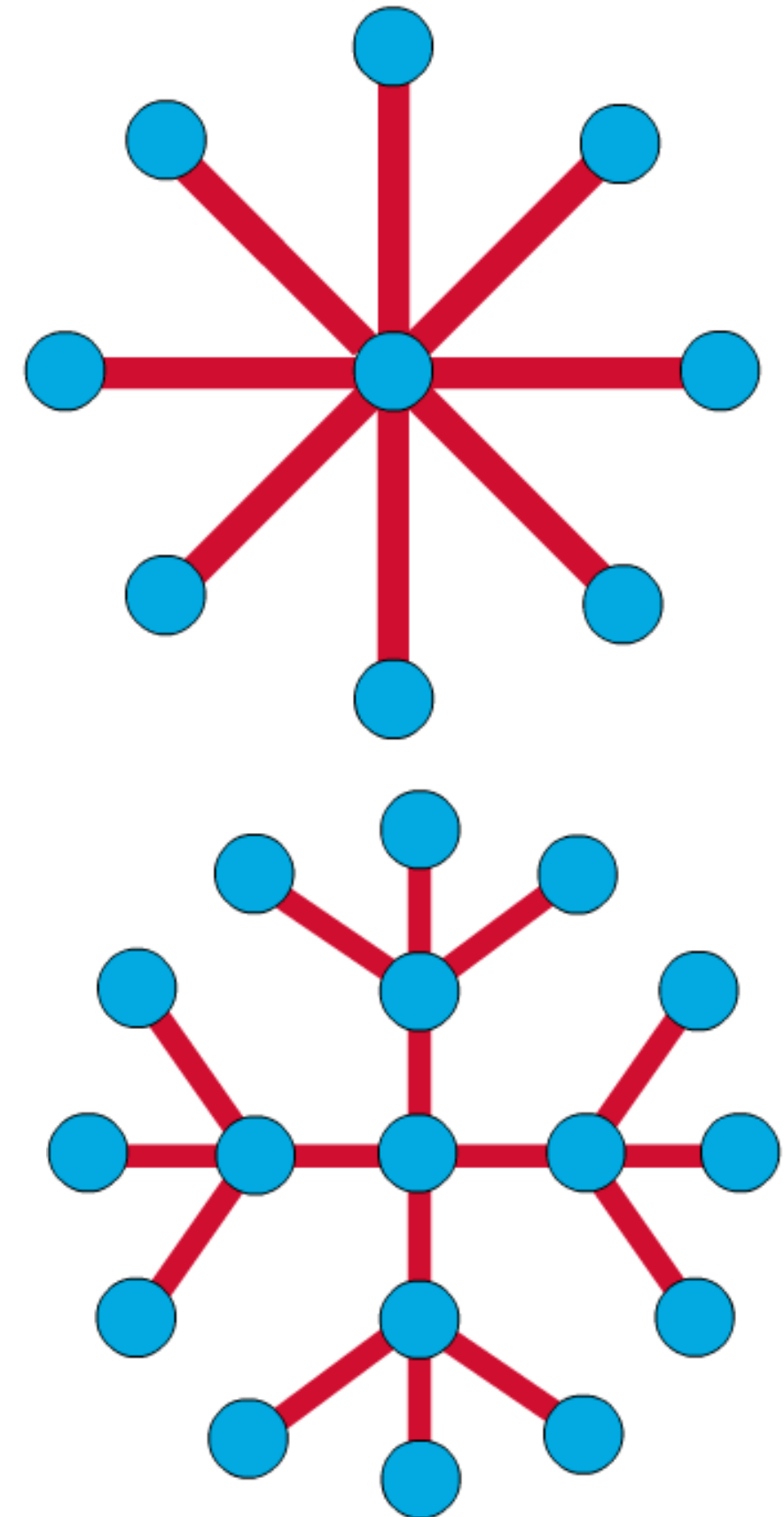
- inexpensive to install as it requires less cabling than star topology and does not require any additional hardware.

3 disadvantages of bus topology

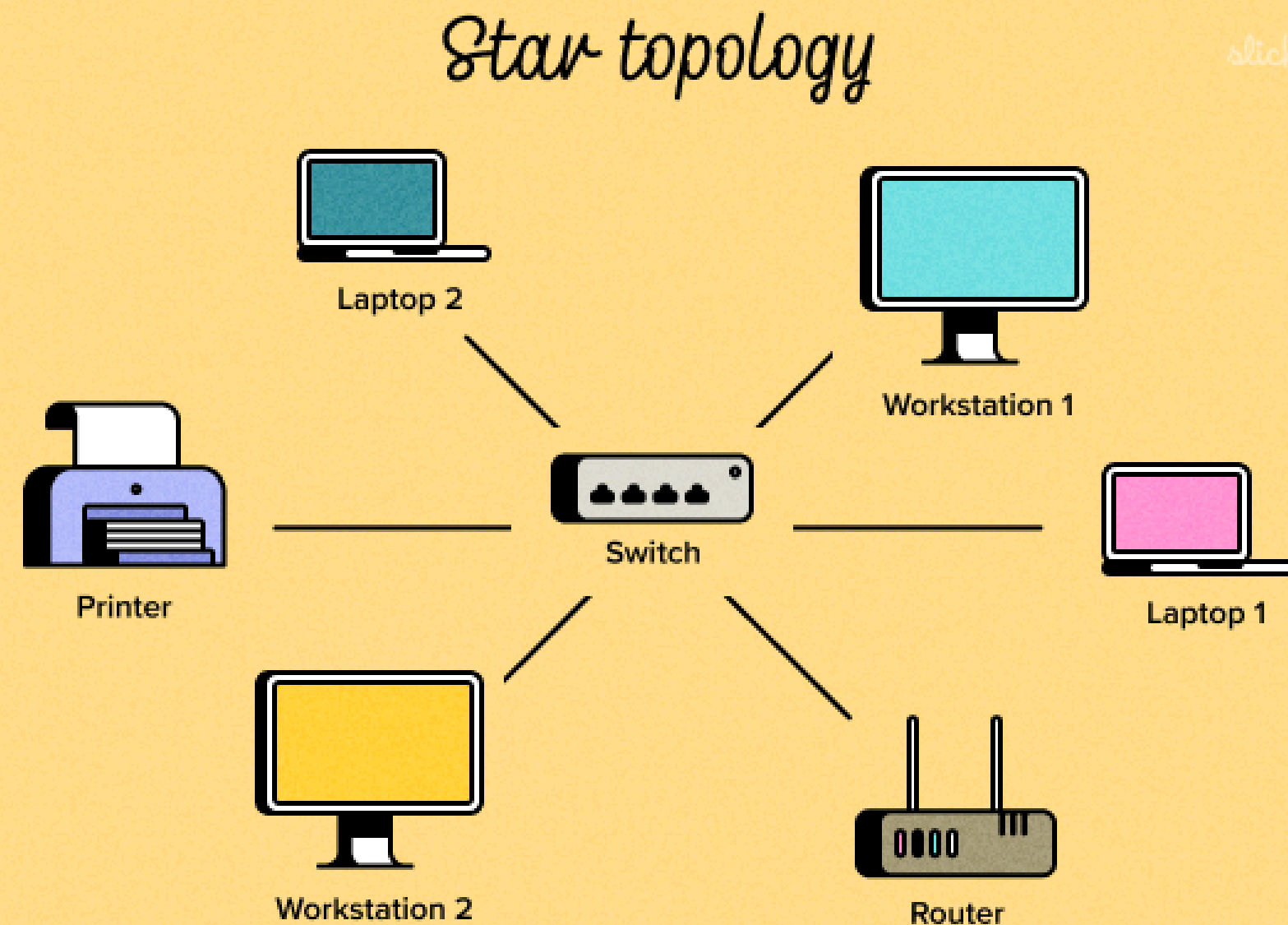
- main cable failure;
- Performance heavy traffic;
- lower security – data transmissions.

Star & Tree Topology

- The star topology is the most commonly used architecture in Ethernet LANs.
- When installed, the star topology resembles spokes in a bicycle wheel.
- Larger networks use the extended star topology also called tree topology. When used with network devices that filter frames or packets, like bridges, switches, and routers, this topology significantly reduces the traffic on the wires by sending packets only to the wires of the destination host.



Star & Tree Topology



Advantages of star topology:

- Easy to troubleshoot
- If one link collapses, only that link is affected. The other links remain active
- No disruption in removal or addition of any devices
- Very easy to manage

Disadvantages of star topology:

- If the hub fails, the entire link will collapse
- Require more cable length in comparison to bus topology
- More expensive than bus topology due to the higher costs of the hubs.

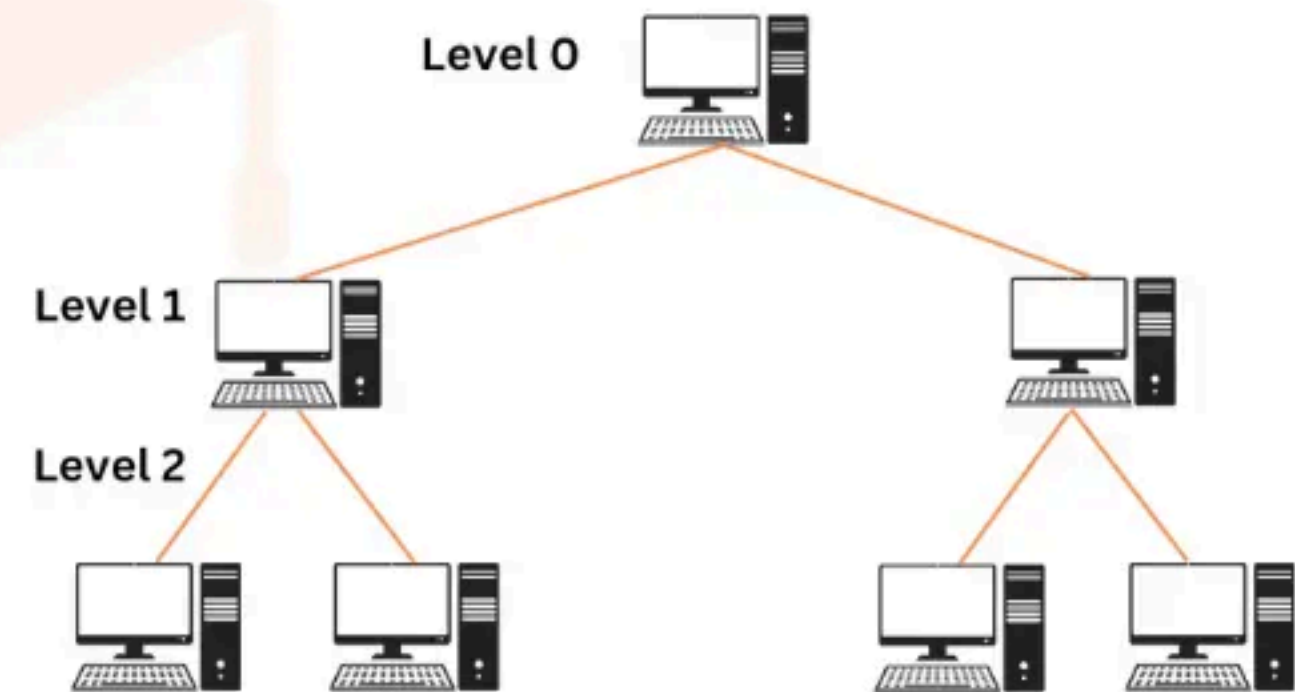
Star & Tree Topology

Advantages

- Easy to Expand
- Organized Structure
- Good for Large Networks
- Supports Both Star and Bus Topology
- Easier Troubleshooting

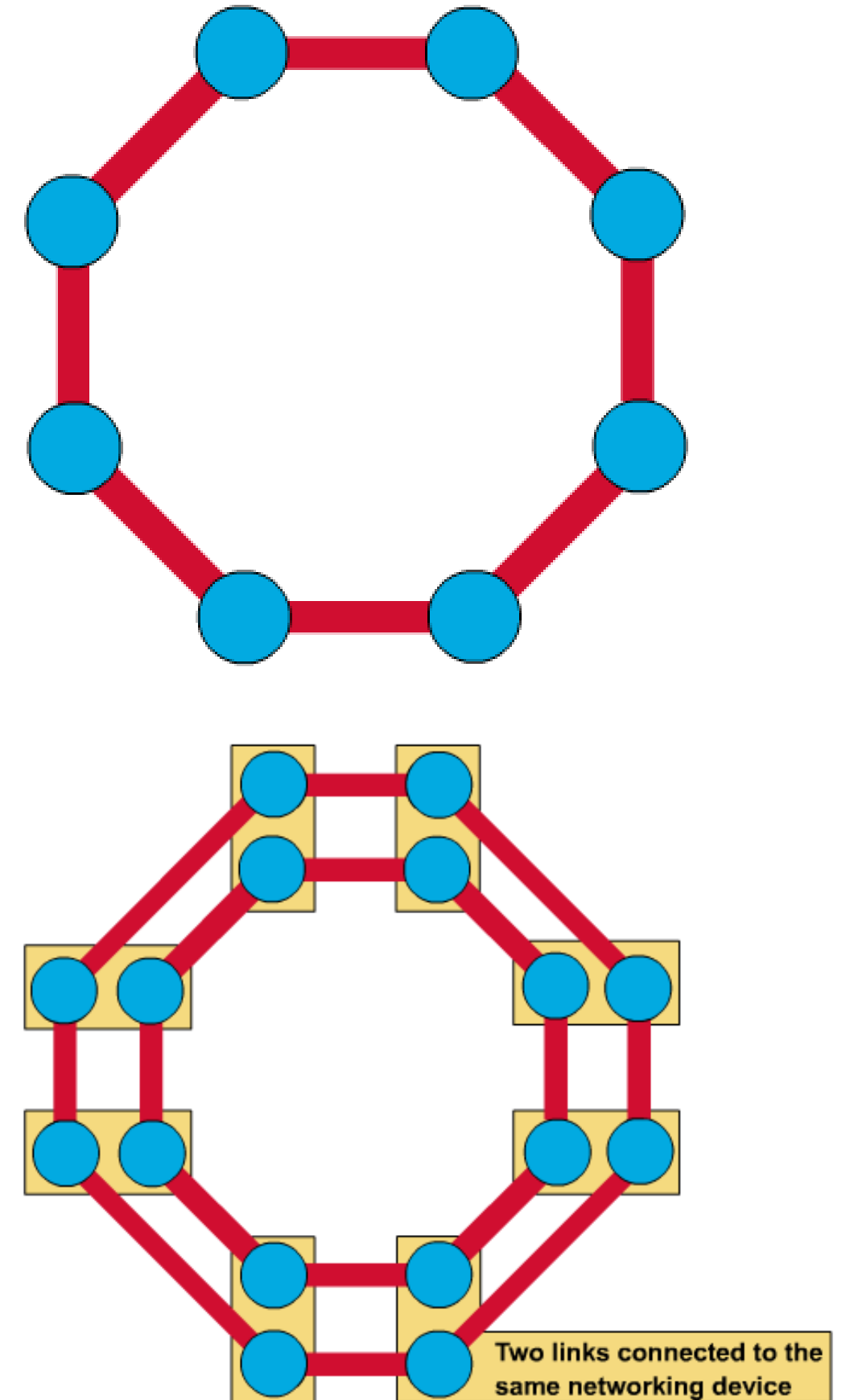
Disadvantages

- Complex and Costly
- Dependency on the Central Hub
- Difficult to Maintain
- Risk of Network Traffic Issues
- Requires Skilled Management



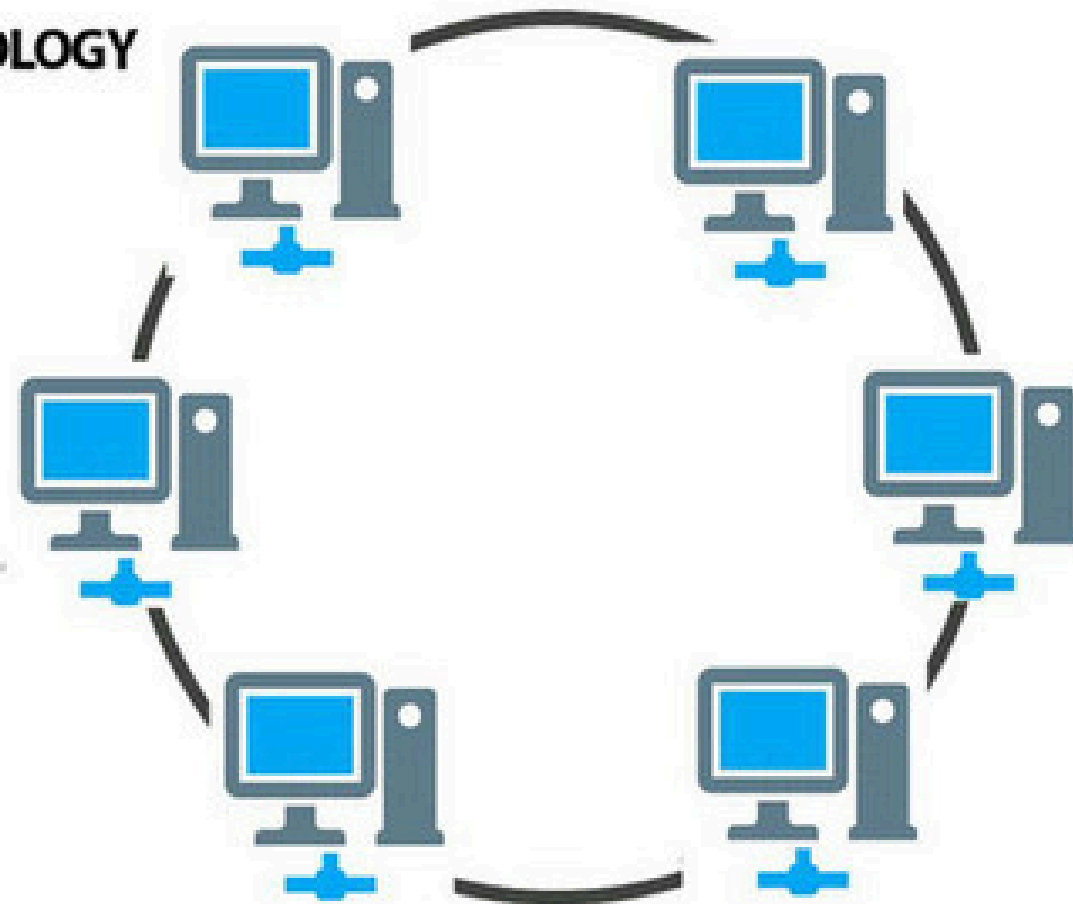
Ring Topology

- A frame travels around the ring, stopping at each node. If a node wants to transmit data, it adds the data as well as the destination address to the frame.
- The frame then continues around the ring until it finds the destination node, which takes the data out of the frame.
 - Single ring – All the devices on the network share a single cable
 - Dual ring – The dual ring topology allows data to be sent in both directions.



Ring Topology

RING TOPOLOGY

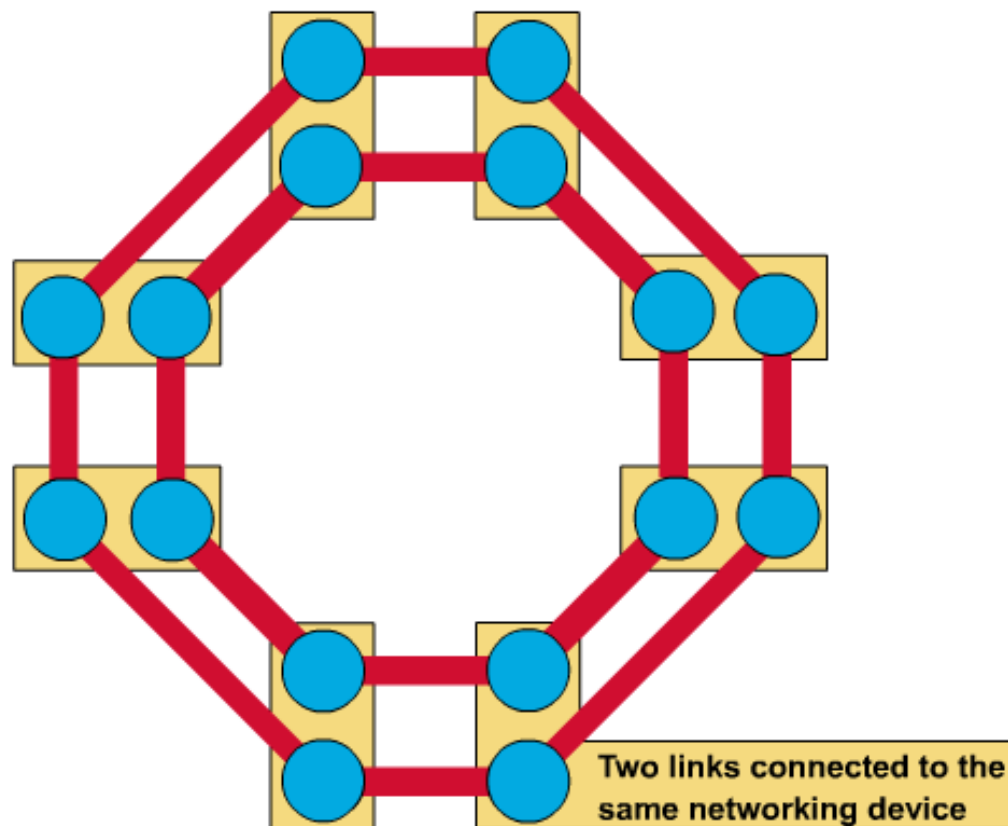


Advantages of ring topology:

- Can handle workload greater than a star topology
- There is no need of network server for the management of the connections
- Cheaper than a star topology as less wiring is required
- A large network can be created by adding a token ring

Disadvantages of ring topology:

- A break in the ring can collapse the entire topology
- Much slower than the Ethernet network
- Any addition or removal of devices affects the entire network



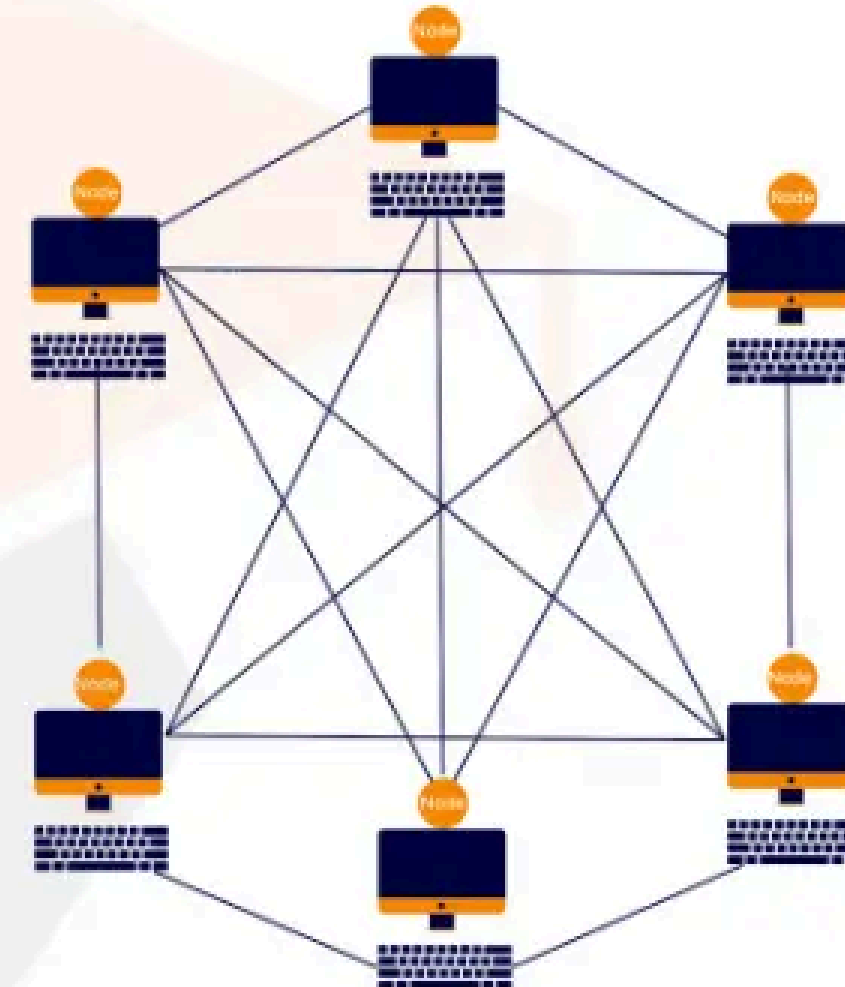
Mesh Topology

Advantages

- Fault Tolerance
- High Data Traffic Handling
- Scalability
- Robustness
- No Centralized Control

Disadvantages

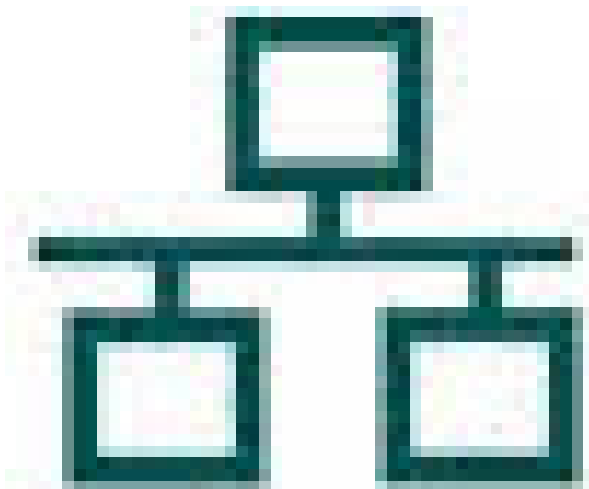
- Costly Implementation
- Complex Installation and Maintenance
- Power Consumption
- Redundant Connections



**MESH
TOPOLOGY**

Communication

Protocols



Communication protocols are sets of rules that control how data is transmitted between devices in embedded systems.

COMMUNICATION PROTOCOLS

Inter system Protocol

USB

(Universal Serial Bus)

UART

(Universal Asynchronous Receiver/Transmitter)

USART

USART (Universal Synchronous/Asynchronous
Receiver/Transmitter)

Intra system Protocol

I2C

(Inter-Integrated Circuit)

SPI

(Serial Peripheral Interface)

CAN

(Controller Area Network)

Communication Software

is designed to connect users through computers or mobile devices, offering global communication options.

- Email
- Instant Messaging
- Voice Over IP (VOIP)
- Video Chat

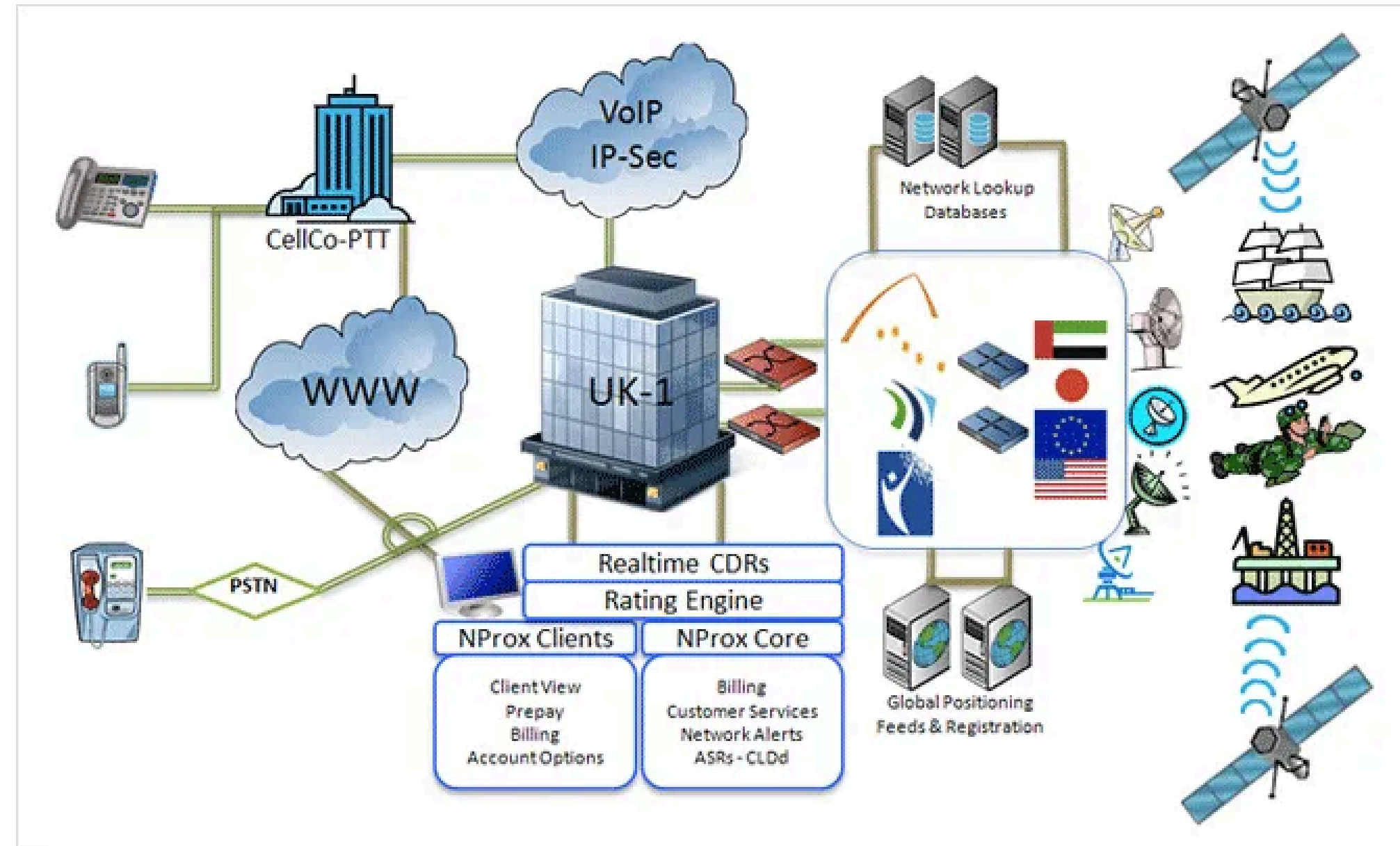




Network services are third-party solutions that manage, design, monitor, secure, and upgrade networking infrastructure.

Key Benefits of Network Services

- No upfront cost
- Better cost prediction
- Improved efficiency
- Higher quality networks
- Enhanced security
- Upgrades & maintenance
- Technical support



Network Architectures



Peer-to-Peer Architecture

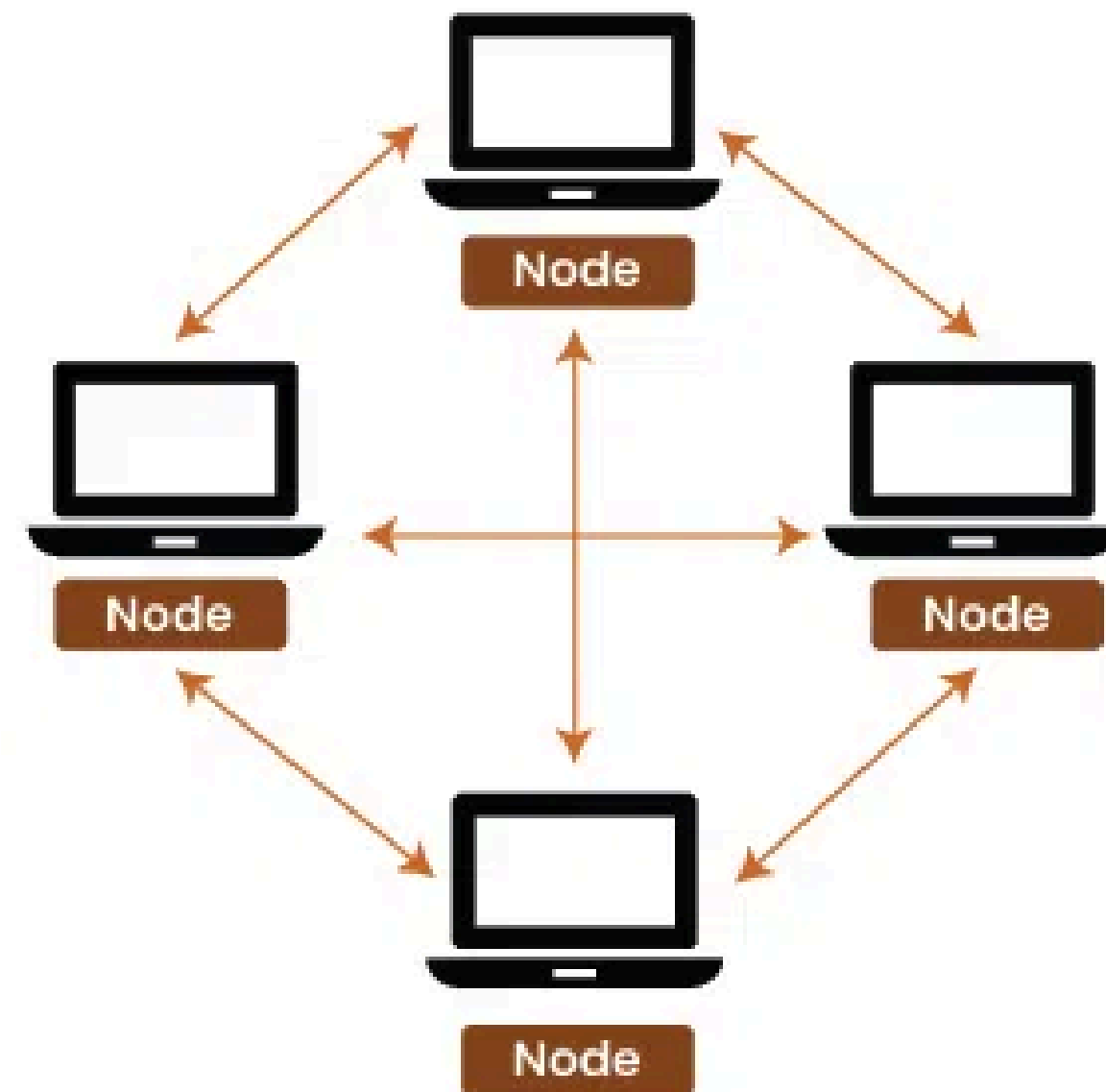
- Equal computers sharing resources
- Easy, low cost, limited security & centralized management



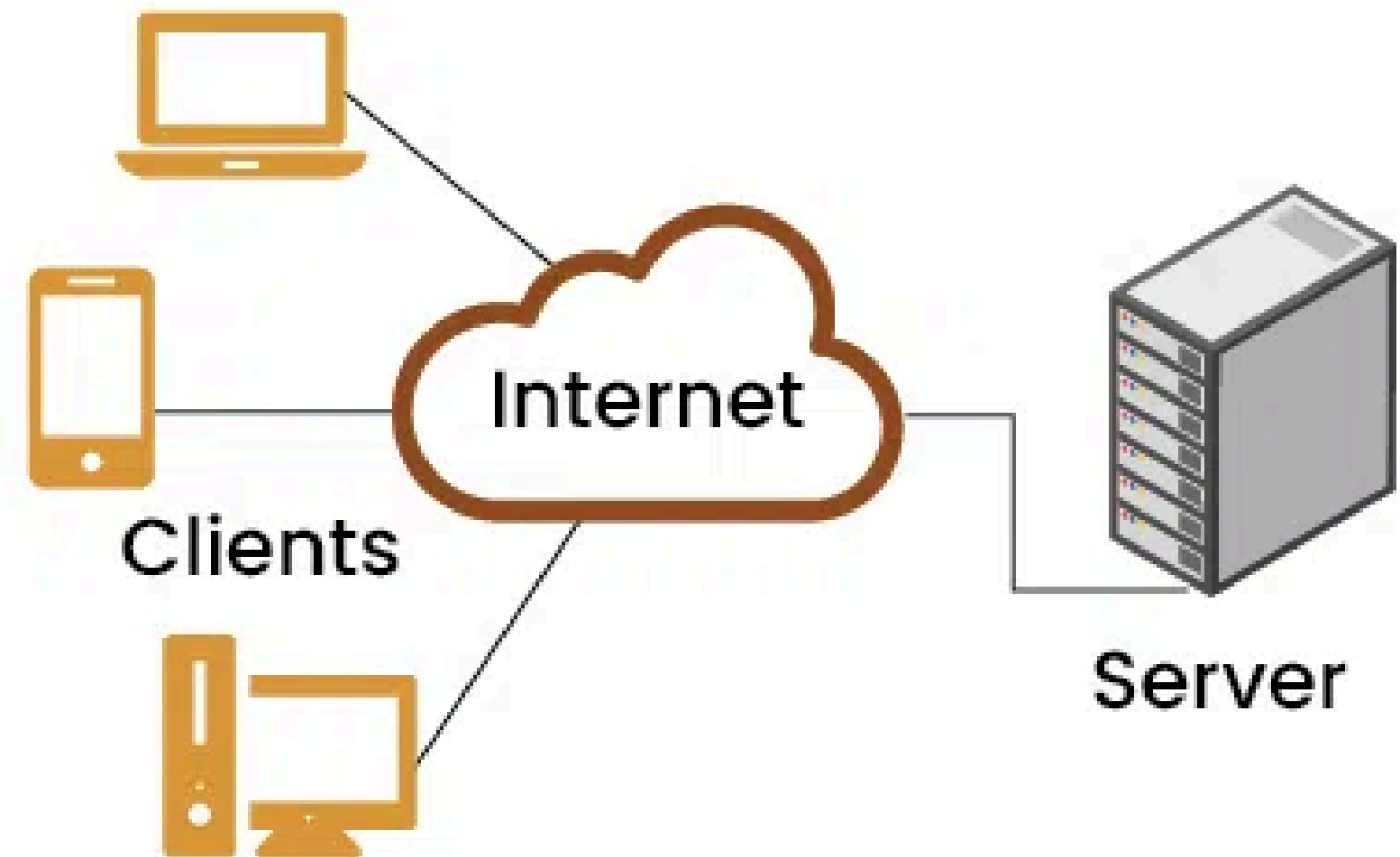
Client/Server Architecture

- Dedicated servers provide services
- Scalable, secure, centralized management
- More complex, higher cost, risk from server failure

Peer to Peer Architecture



Client-Server Architecture



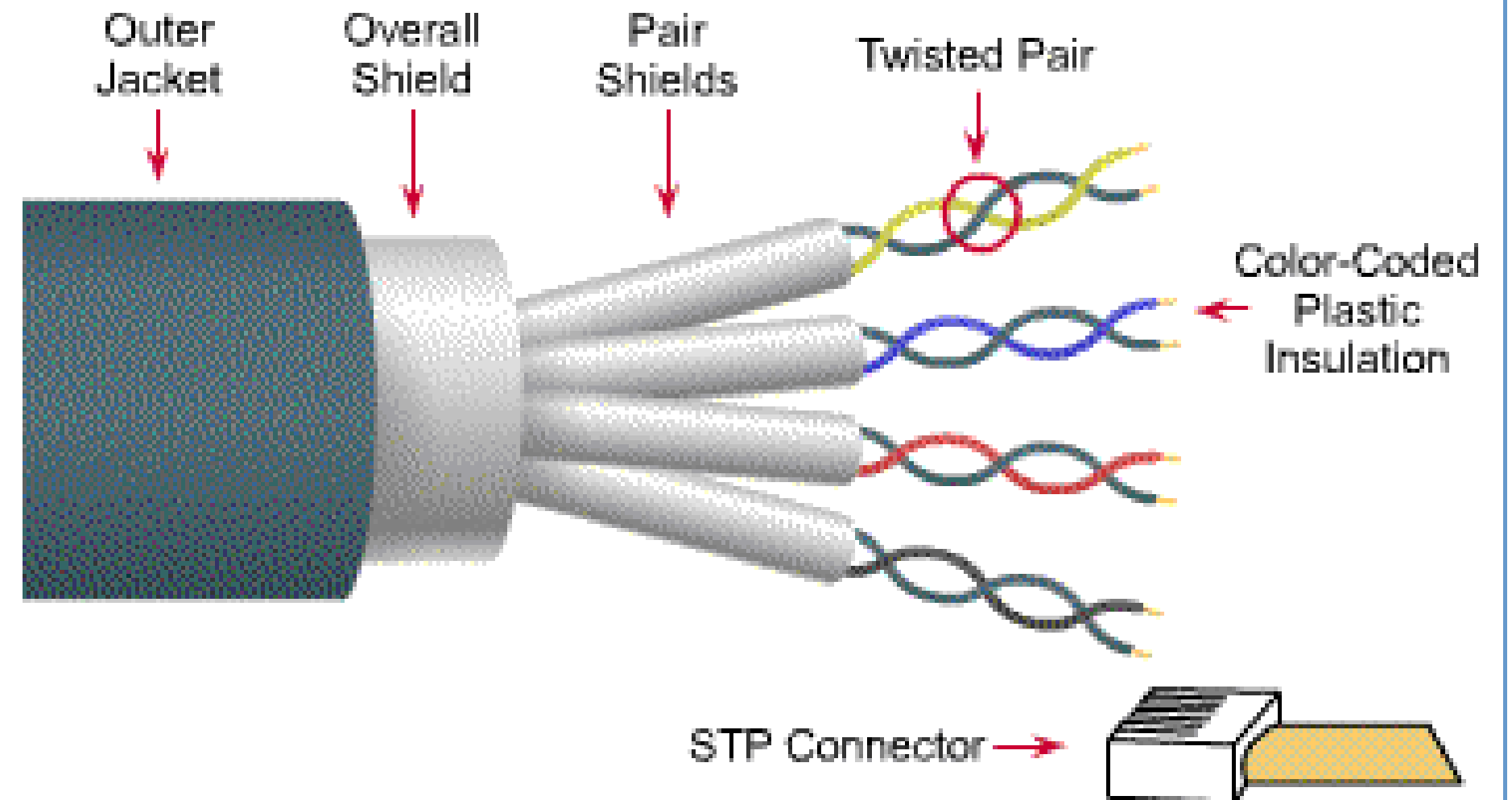


Network Components

- **Physical Media = the wires/waves**
- **Interconnecting Devices = the connectors (routers, switches)**
- **Computers = the users' devices**
- **Networking Software = the managers (OS, firewalls, tools)**
- **Applications = the services we use (email, web, chat, etc.)**

Networking Media

- Networking media can be defined simply as the means by which signals (data) are sent from one computer to another (either by cable or wireless means).



- Speed and throughput: 10-100 Mbps
- Cost per node: Moderately expensive
- Media and connector size: Medium to Large
- Maximum cable length: 100m (short)



Networking Devices

- Hub/Switch
- Router
- AP
- Modem
- Gateway
- Firewall
- Repeater



Layer 2 Switches



Hub



Multi Layer Switches



Firewalls



Modems



Routers



Bridges



Repeater



Thanks